

Experiment 6.2

Convolution of Discrete Time Signals

Program:

% Convolution of Discrete-Time Signals

clc;

clear;

close all;

% Input sequences

x1 = [1 2 3 4];

x2 = [1 1 1];

% Convolution

y = conv(x1,x2);

% Sample indices

n1 = 0:length(x1)-1;

n2 = 0:length(x2)-1;

ny = 0:length(y)-1;

% Display output

disp('Convolution Output:');

disp(y);

% Plotting

figure;

subplot(3,1,1);

stem(n1,x1,'filled');

grid on;

title('Input Sequence x₁(n)');

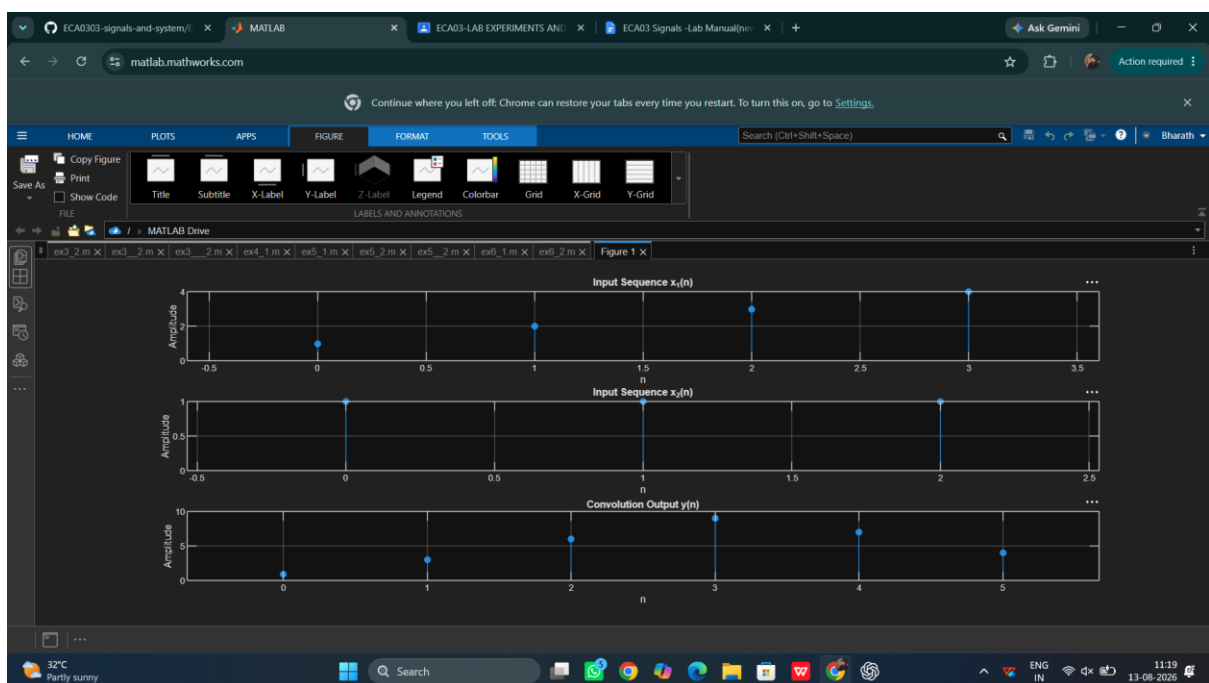
xlabel('n');

```

ylabel('Amplitude');
subplot(3,1,2);
stem(n2,x2,'filled');
grid on;
title('Input Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
stem(ny,y,'filled');
grid on;
title('Convolution Output y(n)');
xlabel('n');
ylabel('Amplitude');

```

Output Waveform:



Result:

The convolution of the discrete-time sequences $x_1(n)=\{1,2,3,4\}$ and $x_2(n)=\{1,1,1\}$ was performed successfully using MATLAB. The output sequence obtained was $y(n)=\{1,3,6,9,7,4\}$ demonstrating the convolution operation for discrete-time signals.